

Artificial Intelligence & Machine Learning

200-Page Study Notes and RAG Evaluation Corpus

Designed for retrieval-augmented generation (RAG) testing, semantic search, chunk retrieval, citation grounding, and question-answer evaluation.

Contents include AI foundations, classical machine learning, deep learning, NLP, computer vision, generative AI, RAG, MLOps, responsible AI, security, and applied use cases.

Page 1: Foundations of Artificial Intelligence

Document ID: AIML-001 | Section: Foundations of Artificial Intelligence

Concept Summary

AI is the field of building systems that perform tasks associated with human intelligence, including perception, reasoning, learning, planning, language understanding, and decision-making.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

foundations, artificial, intelligence, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of foundations of artificial intelligence?
2. What practical issue should be considered when applying foundations of artificial intelligence?
3. How could foundations of artificial intelligence be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 2: History of Artificial Intelligence

Document ID: AIML-002 | Section: History of Artificial Intelligence

Concept Summary

AI developed through several periods: symbolic reasoning, expert systems, statistical learning, deep learning, and foundation models. Progress has been influenced by algorithms, data, computing power, and evaluation methods.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

history, artificial, intelligence, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of history of artificial intelligence?
2. What practical issue should be considered when applying history of artificial intelligence?
3. How could history of artificial intelligence be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 3: Intelligent Agents

Document ID: AIML-003 | Section: Intelligent Agents

Concept Summary

An intelligent agent perceives its environment through sensors and acts through actuators. A rational agent chooses actions that maximize expected performance according to its objective and available information.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

intelligent, agents, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of intelligent agents?
2. What practical issue should be considered when applying intelligent agents?
3. How could intelligent agents be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 4: Problem Solving in AI

Document ID: AIML-004 | Section: Problem Solving in AI

Concept Summary

Many AI tasks can be expressed as a state-space search problem. A state describes a situation, actions create transitions, and a goal test determines whether a solution has been reached.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

problem, solving, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of problem solving in ai?
2. What practical issue should be considered when applying problem solving in ai?
3. How could problem solving in ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 5: Breadth-First Search

Document ID: AIML-005 | Section: Breadth-First Search

Concept Summary

Breadth-first search expands the shallowest unexplored nodes first. With unit step costs, it is complete and returns a shortest path when the branching factor is finite.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

breadth, first, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of breadth-first search?
2. What practical issue should be considered when applying breadth-first search?
3. How could breadth-first search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 6: Depth-First Search

Document ID: AIML-006 | Section: Depth-First Search

Concept Summary

Depth-first search explores one branch as deeply as possible before backtracking. It uses less memory than breadth-first search but may become trapped in deep or infinite branches.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

depth, first, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of depth-first search?
2. What practical issue should be considered when applying depth-first search?
3. How could depth-first search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 7: Uniform-Cost Search

Document ID: AIML-007 | Section: Uniform-Cost Search

Concept Summary

Uniform-cost search expands the node with the lowest path cost. It is optimal when step costs are nonnegative and the queue is ordered by cumulative path cost.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

uniform, cost, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of uniform-cost search?
2. What practical issue should be considered when applying uniform-cost search?
3. How could uniform-cost search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 8: A-Star Search

Document ID: AIML-008 | Section: A-Star Search

Concept Summary

A-star search combines the path cost so far, $g(n)$, with a heuristic estimate, $h(n)$. The evaluation function is $f(n)=g(n)+h(n)$. An admissible heuristic never overestimates the remaining cost.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

star, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of a-star search?
2. What practical issue should be considered when applying a-star search?
3. How could a-star search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 9: Heuristic Functions

Document ID: AIML-009 | Section: Heuristic Functions

Concept Summary

A heuristic guides search toward promising states. Good heuristics reduce computation while preserving solution quality when their properties are understood.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

heuristic, functions, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of heuristic functions?
2. What practical issue should be considered when applying heuristic functions?
3. How could heuristic functions be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 10: Knowledge Representation

Document ID: AIML-010 | Section: Knowledge Representation

Concept Summary

Knowledge representation encodes facts, relationships, rules, and uncertainty in a form that an AI system can use for inference. Common approaches include logic, semantic networks, frames, and ontologies.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

knowledge, representation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of knowledge representation?
2. What practical issue should be considered when applying knowledge representation?
3. How could knowledge representation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 11: Propositional Logic

Document ID: AIML-011 | Section: Propositional Logic

Concept Summary

Propositional logic represents statements that are either true or false. Logical connectives include AND, OR, NOT, implication, and biconditional.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

propositional, logic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of propositional logic?
2. What practical issue should be considered when applying propositional logic?
3. How could propositional logic be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 12: First-Order Logic

Document ID: AIML-012 | Section: First-Order Logic

Concept Summary

First-order logic extends propositional logic with variables, predicates, functions, and quantifiers. It can represent relationships among objects and general rules.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

first, order, logic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of first-order logic?
2. What practical issue should be considered when applying first-order logic?
3. How could first-order logic be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 13: Expert Systems

Document ID: AIML-013 | Section: Expert Systems

Concept Summary

An expert system typically includes a knowledge base and an inference engine. It applies domain rules to known facts and may explain how a conclusion was derived.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

expert, systems, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of expert systems?
2. What practical issue should be considered when applying expert systems?
3. How could expert systems be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 14: Fuzzy Logic

Document ID: AIML-014 | Section: Fuzzy Logic

Concept Summary

Fuzzy logic represents degrees of membership rather than only binary membership. It is useful when concepts such as hot, near, or fast have gradual boundaries.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

fuzzy, logic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of fuzzy logic?
2. What practical issue should be considered when applying fuzzy logic?
3. How could fuzzy logic be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 15: Probability in AI

Document ID: AIML-015 | Section: Probability in AI

Concept Summary

Probability provides a mathematical language for uncertainty. Conditional probability, Bayes' theorem, and probability distributions are central to probabilistic reasoning.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

probability, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of probability in ai?
2. What practical issue should be considered when applying probability in ai?
3. How could probability in ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 16: Bayesian Networks

Document ID: AIML-016 | Section: Bayesian Networks

Concept Summary

A Bayesian network is a directed acyclic graph whose nodes represent random variables and whose edges represent conditional dependencies. It factorizes a joint probability distribution.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

bayesian, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of bayesian networks?
2. What practical issue should be considered when applying bayesian networks?
3. How could bayesian networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 17: Decision Trees

Document ID: AIML-017 | Section: Decision Trees

Concept Summary

A decision tree predicts a target by applying a sequence of feature-based tests. Splits may be selected using information gain, gain ratio, or Gini impurity.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

decision, trees, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of decision trees?
2. What practical issue should be considered when applying decision trees?
3. How could decision trees be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 18: Machine Learning Overview

Document ID: AIML-018 | Section: Machine Learning Overview

Concept Summary

Machine learning enables a system to learn patterns from examples rather than relying entirely on manually written rules. The learning process includes data preparation, training, validation, and testing.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

machine, learning, overview, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of machine learning overview?
2. What practical issue should be considered when applying machine learning overview?
3. How could machine learning overview be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 19: Supervised Learning

Document ID: AIML-019 | Section: Supervised Learning

Concept Summary

Supervised learning uses labeled examples. The model learns a mapping from input variables to target outputs and is evaluated using metrics suitable for the task.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

supervised, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of supervised learning?
2. What practical issue should be considered when applying supervised learning?
3. How could supervised learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 20: Unsupervised Learning

Document ID: AIML-020 | Section: Unsupervised Learning

Concept Summary

Unsupervised learning identifies structure in unlabeled data. Clustering, dimensionality reduction, and density estimation are common examples.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

unsupervised, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of unsupervised learning?
2. What practical issue should be considered when applying unsupervised learning?
3. How could unsupervised learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 21: Semi-Supervised Learning

Document ID: AIML-021 | Section: Semi-Supervised Learning

Concept Summary

Semi-supervised learning combines a small labeled dataset with a larger unlabeled dataset. It can be useful when labeling is expensive but raw data is abundant.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

semi, supervised, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of semi-supervised learning?
2. What practical issue should be considered when applying semi-supervised learning?
3. How could semi-supervised learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 22: Reinforcement Learning

Document ID: AIML-022 | Section: Reinforcement Learning

Concept Summary

Reinforcement learning trains an agent through interactions with an environment. The agent receives rewards and learns a policy for selecting actions.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

reinforcement, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of reinforcement learning?
2. What practical issue should be considered when applying reinforcement learning?
3. How could reinforcement learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 23: Regression

Document ID: AIML-023 | Section: Regression

Concept Summary

Regression predicts a numerical value. Linear regression assumes a linear relationship between features and a target, while nonlinear methods can model more complex patterns.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

regression, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of regression?
2. What practical issue should be considered when applying regression?
3. How could regression be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 24: Linear Regression

Document ID: AIML-024 | Section: Linear Regression

Concept Summary

Linear regression estimates coefficients that minimize a loss, commonly mean squared error. Its assumptions should be examined before interpreting coefficients.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

linear, regression, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of linear regression?
2. What practical issue should be considered when applying linear regression?
3. How could linear regression be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 25: Polynomial Regression

Document ID: AIML-025 | Section: Polynomial Regression

Concept Summary

Polynomial regression adds powers or combinations of an input feature to represent curved relationships. Regularization may be needed to control overfitting.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

polynomial, regression, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of polynomial regression?
2. What practical issue should be considered when applying polynomial regression?
3. How could polynomial regression be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 26: Logistic Regression

Document ID: AIML-026 | Section: Logistic Regression

Concept Summary

Logistic regression models the probability of a class using the logistic function. Despite its name, it is commonly used for classification.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

logistic, regression, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of logistic regression?
2. What practical issue should be considered when applying logistic regression?
3. How could logistic regression be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 27: K-Nearest Neighbors

Document ID: AIML-027 | Section: K-Nearest Neighbors

Concept Summary

K-nearest neighbors predicts using the labels or values of nearby training examples. Feature scaling is important because distance calculations are sensitive to measurement units.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

nearest, neighbors, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of k-nearest neighbors?
2. What practical issue should be considered when applying k-nearest neighbors?
3. How could k-nearest neighbors be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 28: Support Vector Machines

Document ID: AIML-028 | Section: Support Vector Machines

Concept Summary

Support vector machines seek a decision boundary with a large margin. Kernel functions can map inputs into spaces where separation is easier.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

support, vector, machines, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of support vector machines?
2. What practical issue should be considered when applying support vector machines?
3. How could support vector machines be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 29: Naive Bayes

Document ID: AIML-029 | Section: Naive Bayes

Concept Summary

Naive Bayes applies Bayes' theorem while assuming conditional independence among features given the class. It is efficient and often effective for text classification.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

naive, bayes, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of naive bayes?
2. What practical issue should be considered when applying naive bayes?
3. How could naive bayes be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 30: Random Forest

Document ID: AIML-030 | Section: Random Forest

Concept Summary

A random forest combines many decision trees trained with randomized samples and feature selections. Aggregation can reduce variance and improve generalization.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

random, forest, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of random forest?
2. What practical issue should be considered when applying random forest?
3. How could random forest be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 31: Gradient Boosting

Document ID: AIML-031 | Section: Gradient Boosting

Concept Summary

Gradient boosting builds an additive model sequentially. Each new learner attempts to reduce the errors of the current ensemble.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

gradient, boosting, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of gradient boosting?
2. What practical issue should be considered when applying gradient boosting?
3. How could gradient boosting be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 32: XGBoost Concepts

Document ID: AIML-032 | Section: XGBoost Concepts

Concept Summary

Extreme gradient boosting uses regularized tree boosting with efficient optimization and engineering improvements. Important settings include learning rate, tree depth, and number of estimators.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

xgboost, concepts, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of xgboost concepts?
2. What practical issue should be considered when applying xgboost concepts?
3. How could xgboost concepts be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 33: Clustering

Document ID: AIML-033 | Section: Clustering

Concept Summary

Clustering groups examples so that members of the same group are more similar under a chosen distance or similarity measure.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

clustering, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of clustering?
2. What practical issue should be considered when applying clustering?
3. How could clustering be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 34: K-Means

Document ID: AIML-034 | Section: K-Means

Concept Summary

K-means alternates between assigning points to the nearest centroid and updating centroids. The number of clusters must be selected or justified.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

means, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of k-means?
2. What practical issue should be considered when applying k-means?
3. How could k-means be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 35: Hierarchical Clustering

Document ID: AIML-035 | Section: Hierarchical Clustering

Concept Summary

Hierarchical clustering creates a nested structure of groups. Agglomerative methods start with individual points and repeatedly merge clusters.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

hierarchical, clustering, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of hierarchical clustering?
2. What practical issue should be considered when applying hierarchical clustering?
3. How could hierarchical clustering be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 36: DBSCAN

Document ID: AIML-036 | Section: DBSCAN

Concept Summary

DBSCAN groups dense regions and marks sparse points as noise. It can identify irregularly shaped clusters and does not require specifying the number of clusters.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

dbscan, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of dbscan?
2. What practical issue should be considered when applying dbscan?
3. How could dbscan be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 37: Principal Component Analysis

Document ID: AIML-037 | Section: Principal Component Analysis

Concept Summary

PCA transforms correlated variables into orthogonal components ordered by explained variance. It is often used for compression, visualization, and noise reduction.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

principal, component, analysis, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of principal component analysis?
2. What practical issue should be considered when applying principal component analysis?
3. How could principal component analysis be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 38: Feature Engineering

Document ID: AIML-038 | Section: Feature Engineering

Concept Summary

Feature engineering converts raw information into useful model inputs. Examples include ratios, counts, date components, text statistics, and domain-specific transformations.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

feature, engineering, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of feature engineering?
2. What practical issue should be considered when applying feature engineering?
3. How could feature engineering be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 39: Feature Scaling

Document ID: AIML-039 | Section: Feature Scaling

Concept Summary

Scaling puts numerical features on comparable ranges. Standardization subtracts the mean and divides by standard deviation, while min-max scaling maps values to a selected interval.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

feature, scaling, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of feature scaling?
2. What practical issue should be considered when applying feature scaling?
3. How could feature scaling be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 40: Missing Data

Document ID: AIML-040 | Section: Missing Data

Concept Summary

Missing values can be handled through deletion, simple imputation, model-based imputation, or explicit missing indicators. The method should respect the data-generating process.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

missing, data, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of missing data?
2. What practical issue should be considered when applying missing data?
3. How could missing data be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 41: Outliers

Document ID: AIML-041 | Section: Outliers

Concept Summary

Outliers are observations that differ substantially from the rest of the data. They may be errors, rare valid cases, or evidence of a different population.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

outliers, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of outliers?
2. What practical issue should be considered when applying outliers?
3. How could outliers be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 42: Data Leakage

Document ID: AIML-042 | Section: Data Leakage

Concept Summary

Data leakage occurs when information unavailable at prediction time influences training. Leakage produces overly optimistic evaluation and poor real-world performance.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

data, leakage, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of data leakage?
2. What practical issue should be considered when applying data leakage?
3. How could data leakage be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 43: Train Validation Test Split

Document ID: AIML-043 | Section: Train Validation Test Split

Concept Summary

A dataset is commonly divided into training, validation, and test sets. The test set should remain isolated until final evaluation.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

train, validation, test, split, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of train validation test split?
2. What practical issue should be considered when applying train validation test split?
3. How could train validation test split be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 44: Cross Validation

Document ID: AIML-044 | Section: Cross Validation

Concept Summary

Cross-validation repeatedly partitions the training data to estimate generalization. K-fold cross-validation divides data into k folds and rotates the validation fold.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

cross, validation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of cross validation?
2. What practical issue should be considered when applying cross validation?
3. How could cross validation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 45: Overfitting

Document ID: AIML-045 | Section: Overfitting

Concept Summary

Overfitting happens when a model learns noise or accidental details in training data. Symptoms include low training error and substantially higher validation error.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

overfitting, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of overfitting?
2. What practical issue should be considered when applying overfitting?
3. How could overfitting be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 46: Underfitting

Document ID: AIML-046 | Section: Underfitting

Concept Summary

Underfitting occurs when a model is too simple to capture important patterns. It can be addressed through better features, increased capacity, or reduced regularization.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

underfitting, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of underfitting?
2. What practical issue should be considered when applying underfitting?
3. How could underfitting be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 47: Bias Variance Tradeoff

Document ID: AIML-047 | Section: Bias Variance Tradeoff

Concept Summary

Bias represents systematic error from restrictive assumptions, while variance represents sensitivity to training data. Model selection balances both sources of error.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

bias, variance, tradeoff, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of bias variance tradeoff?
2. What practical issue should be considered when applying bias variance tradeoff?
3. How could bias variance tradeoff be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 48: Regularization

Document ID: AIML-048 | Section: Regularization

Concept Summary

Regularization discourages overly complex models. L1 regularization can encourage sparse coefficients, while L2 regularization shrinks coefficients toward zero.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

regularization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of regularization?
2. What practical issue should be considered when applying regularization?
3. How could regularization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 49: Hyperparameter Tuning

Document ID: AIML-049 | Section: Hyperparameter Tuning

Concept Summary

Hyperparameters are chosen before or during training, such as tree depth, learning rate, and regularization strength. Grid search, random search, and Bayesian optimization are common methods.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

hyperparameter, tuning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of hyperparameter tuning?
2. What practical issue should be considered when applying hyperparameter tuning?
3. How could hyperparameter tuning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 50: Evaluation Metrics

Document ID: AIML-050 | Section: Evaluation Metrics

Concept Summary

Metrics should reflect the task and its costs. Classification metrics include accuracy, precision, recall, F1 score, and ROC-AUC; regression metrics include MAE, MSE, RMSE, and R-squared.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

evaluation, metrics, model, data, deployment.

RAG Test Questions

1. What is the main idea of evaluation metrics?
2. What practical issue should be considered when applying evaluation metrics?
3. How could evaluation metrics be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 51: Confusion Matrix

Document ID: AIML-051 | Section: Confusion Matrix

Concept Summary

A confusion matrix counts true positives, true negatives, false positives, and false negatives. It helps identify the types of errors a classifier makes.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

confusion, matrix, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of confusion matrix?
2. What practical issue should be considered when applying confusion matrix?
3. How could confusion matrix be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 52: Class Imbalance

Document ID: AIML-052 | Section: Class Imbalance

Concept Summary

Class imbalance occurs when one class is much more frequent than another. Options include class weighting, resampling, threshold adjustment, and suitable metrics.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

class, imbalance, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of class imbalance?
2. What practical issue should be considered when applying class imbalance?
3. How could class imbalance be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 53: Calibration

Document ID: AIML-053 | Section: Calibration

Concept Summary

A calibrated classifier produces probabilities that correspond to observed frequencies. Calibration curves and reliability diagrams help evaluate probability quality.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

calibration, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of calibration?
2. What practical issue should be considered when applying calibration?
3. How could calibration be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 54: Explainable AI

Document ID: AIML-054 | Section: Explainable AI

Concept Summary

Explainable AI aims to make model behavior understandable to users. Feature importance, partial dependence, counterfactuals, and local explanations provide different views.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

explainable, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of explainable ai?
2. What practical issue should be considered when applying explainable ai?
3. How could explainable ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 55: Fairness in AI

Document ID: AIML-055 | Section: Fairness in AI

Concept Summary

Fairness analysis examines whether model outcomes differ across relevant groups. The appropriate fairness definition depends on the application, stakeholders, and legal context.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

fairness, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of fairness in ai?
2. What practical issue should be considered when applying fairness in ai?
3. How could fairness in ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 56: Privacy in Machine Learning

Document ID: AIML-056 | Section: Privacy in Machine Learning

Concept Summary

Privacy-preserving approaches include data minimization, access control, anonymization, federated learning, and differential privacy. Each method has trade-offs.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

privacy, machine, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of privacy in machine learning?
2. What practical issue should be considered when applying privacy in machine learning?
3. How could privacy in machine learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 57: AI Safety

Document ID: AIML-057 | Section: AI Safety

Concept Summary

AI safety includes robustness, monitoring, misuse prevention, human oversight, and fail-safe behavior. Safety requirements should be defined before deployment.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

safety, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai safety?
2. What practical issue should be considered when applying ai safety?
3. How could ai safety be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 58: Neural Networks

Document ID: AIML-058 | Section: Neural Networks

Concept Summary

A neural network is composed of layers of weighted transformations and nonlinear activation functions. Training adjusts weights to reduce a loss function.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

neural, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of neural networks?
2. What practical issue should be considered when applying neural networks?
3. How could neural networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 59: Perceptron

Document ID: AIML-059 | Section: Perceptron

Concept Summary

The perceptron computes a weighted sum and applies a threshold. It can represent linearly separable binary classification problems.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

perceptron, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of perceptron?
2. What practical issue should be considered when applying perceptron?
3. How could perceptron be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 60: Backpropagation

Document ID: AIML-060 | Section: Backpropagation

Concept Summary

Backpropagation applies the chain rule to calculate gradients of the loss with respect to network parameters. An optimizer uses these gradients to update weights.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

backpropagation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of backpropagation?
2. What practical issue should be considered when applying backpropagation?
3. How could backpropagation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 61: Activation Functions

Document ID: AIML-061 | Section: Activation Functions

Concept Summary

Activation functions introduce nonlinearity. ReLU is common in hidden layers, while sigmoid and softmax are frequently used for output probabilities.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

activation, functions, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of activation functions?
2. What practical issue should be considered when applying activation functions?
3. How could activation functions be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 62: Loss Functions

Document ID: AIML-062 | Section: Loss Functions

Concept Summary

A loss function measures the discrepancy between predictions and targets. Mean squared error is common for regression, while cross-entropy is common for classification.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

loss, functions, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of loss functions?
2. What practical issue should be considered when applying loss functions?
3. How could loss functions be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 63: Optimization

Document ID: AIML-063 | Section: Optimization

Concept Summary

Optimization algorithms update model parameters to minimize loss. Gradient descent, stochastic gradient descent, momentum, and adaptive methods are widely used.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

optimization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of optimization?
2. What practical issue should be considered when applying optimization?
3. How could optimization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 64: Batch Size and Epochs

Document ID: AIML-064 | Section: Batch Size and Epochs

Concept Summary

Batch size determines how many examples contribute to one update. An epoch represents one pass through the training dataset.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

batch, size, epochs, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of batch size and epochs?
2. What practical issue should be considered when applying batch size and epochs?
3. How could batch size and epochs be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 65: Dropout

Document ID: AIML-065 | Section: Dropout

Concept Summary

Dropout randomly disables selected activations during training. It acts as a regularizer and can reduce co-adaptation among neurons.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

dropout, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of dropout?
2. What practical issue should be considered when applying dropout?
3. How could dropout be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 66: Batch Normalization

Document ID: AIML-066 | Section: Batch Normalization

Concept Summary

Batch normalization normalizes intermediate activations using batch statistics during training and learned parameters. It can improve optimization stability.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

batch, normalization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of batch normalization?
2. What practical issue should be considered when applying batch normalization?
3. How could batch normalization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 67: Convolutional Neural Networks

Document ID: AIML-067 | Section: Convolutional Neural Networks

Concept Summary

CNNs use local filters and shared weights to learn spatial patterns. They are widely used for image and video processing.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

convolutional, neural, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of convolutional neural networks?
2. What practical issue should be considered when applying convolutional neural networks?
3. How could convolutional neural networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 68: Pooling Layers

Document ID: AIML-068 | Section: Pooling Layers

Concept Summary

Pooling summarizes local neighborhoods, reducing spatial dimensions and sometimes improving translation tolerance.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

pooling, layers, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of pooling layers?
2. What practical issue should be considered when applying pooling layers?
3. How could pooling layers be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 69: Image Classification

Document ID: AIML-069 | Section: Image Classification

Concept Summary

Image classification assigns one or more labels to an image. A typical pipeline includes preprocessing, augmentation, feature extraction, and classification.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

image, classification, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of image classification?
2. What practical issue should be considered when applying image classification?
3. How could image classification be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 70: Object Detection

Document ID: AIML-070 | Section: Object Detection

Concept Summary

Object detection identifies object categories and their locations, often represented by bounding boxes. Detection quality depends on localization and classification.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

object, detection, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of object detection?
2. What practical issue should be considered when applying object detection?
3. How could object detection be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 71: Image Segmentation

Document ID: AIML-071 | Section: Image Segmentation

Concept Summary

Segmentation assigns a class to each pixel or region. Semantic segmentation labels categories, while instance segmentation distinguishes individual objects.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

image, segmentation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of image segmentation?
2. What practical issue should be considered when applying image segmentation?
3. How could image segmentation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 72: Data Augmentation

Document ID: AIML-072 | Section: Data Augmentation

Concept Summary

Data augmentation creates varied training examples through transformations such as cropping, flipping, rotation, noise, and color changes.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

data, augmentation, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of data augmentation?
2. What practical issue should be considered when applying data augmentation?
3. How could data augmentation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 73: Recurrent Neural Networks

Document ID: AIML-073 | Section: Recurrent Neural Networks

Concept Summary

RNNs process sequences while maintaining a hidden state. They can model temporal dependencies but may struggle with long-range gradients.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

recurrent, neural, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of recurrent neural networks?
2. What practical issue should be considered when applying recurrent neural networks?
3. How could recurrent neural networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 74: LSTM Networks

Document ID: AIML-074 | Section: LSTM Networks

Concept Summary

Long short-term memory networks use gates to regulate information flow through a cell state. They were designed to improve learning of long dependencies.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

Lstm, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of lstm networks?
2. What practical issue should be considered when applying lstm networks?
3. How could lstm networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 75: GRU Networks

Document ID: AIML-075 | Section: GRU Networks

Concept Summary

Gated recurrent units use update and reset gates with fewer components than LSTMs. They can provide a practical alternative for sequence modeling.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of gru networks?
2. What practical issue should be considered when applying gru networks?
3. How could gru networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 76: Attention Mechanism

Document ID: AIML-076 | Section: Attention Mechanism

Concept Summary

Attention computes weighted combinations of representations based on relevance. It allows a model to focus on different parts of an input.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

attention, mechanism, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of attention mechanism?
2. What practical issue should be considered when applying attention mechanism?
3. How could attention mechanism be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 77: Transformers

Document ID: AIML-077 | Section: Transformers

Concept Summary

Transformers rely primarily on attention and feed-forward layers rather than recurrence. They support parallel processing and have become central to language and multimodal models.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

transformers, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of transformers?
2. What practical issue should be considered when applying transformers?
3. How could transformers be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 78: Self-Attention

Document ID: AIML-078 | Section: Self-Attention

Concept Summary

Self-attention relates tokens within the same sequence. Queries, keys, and values are used to calculate attention weights.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

self, attention, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of self-attention?
2. What practical issue should be considered when applying self-attention?
3. How could self-attention be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 79: Embeddings

Document ID: AIML-079 | Section: Embeddings

Concept Summary

An embedding maps an object such as a word, sentence, image, or user into a numerical vector space. Similar objects may have nearby vectors.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

embeddings, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of embeddings?
2. What practical issue should be considered when applying embeddings?
3. How could embeddings be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 80: Natural Language Processing

Document ID: AIML-080 | Section: Natural Language Processing

Concept Summary

NLP applies computational methods to human language. Tasks include classification, information extraction, translation, summarization, and question answering.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

natural, language, processing, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of natural language processing?
2. What practical issue should be considered when applying natural language processing?
3. How could natural language processing be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 81: Tokenization

Document ID: AIML-081 | Section: Tokenization

Concept Summary

Tokenization splits text into units used by a model. Units may be words, subwords, characters, or other segments.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

tokenization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of tokenization?
2. What practical issue should be considered when applying tokenization?
3. How could tokenization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 82: Text Classification

Document ID: AIML-082 | Section: Text Classification

Concept Summary

Text classification assigns categories to documents or messages. Preprocessing, representation, class balance, and evaluation influence performance.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

text, classification, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of text classification?
2. What practical issue should be considered when applying text classification?
3. How could text classification be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 83: Named Entity Recognition

Document ID: AIML-083 | Section: Named Entity Recognition

Concept Summary

Named entity recognition identifies spans referring to entities such as people, organizations, locations, dates, and products.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

named, entity, recognition, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of named entity recognition?
2. What practical issue should be considered when applying named entity recognition?
3. How could named entity recognition be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 84: Sentiment Analysis

Document ID: AIML-084 | Section: Sentiment Analysis

Concept Summary

Sentiment analysis estimates opinions or emotional polarity in text. Domain-specific language, sarcasm, and context can make the task difficult.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

sentiment, analysis, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of sentiment analysis?
2. What practical issue should be considered when applying sentiment analysis?
3. How could sentiment analysis be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 85: Text Summarization

Document ID: AIML-085 | Section: Text Summarization

Concept Summary

Summarization produces a shorter representation of source material. Extractive methods select source passages, while abstractive methods generate new wording.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

text, summarization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of text summarization?
2. What practical issue should be considered when applying text summarization?
3. How could text summarization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 86: Machine Translation

Document ID: AIML-086 | Section: Machine Translation

Concept Summary

Machine translation converts text between languages. Evaluation may use automatic metrics, human review, and task-specific adequacy criteria.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

machine, translation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of machine translation?
2. What practical issue should be considered when applying machine translation?
3. How could machine translation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 87: Question Answering

Document ID: AIML-087 | Section: Question Answering

Concept Summary

Question answering systems return answers to questions using a model, a knowledge source, or both. Retrieval grounding can reduce unsupported responses.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

question, answering, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of question answering?
2. What practical issue should be considered when applying question answering?
3. How could question answering be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 88: Large Language Models

Document ID: AIML-088 | Section: Large Language Models

Concept Summary

Large language models learn statistical patterns from large text collections. They can generate, transform, classify, and reason over language, but may produce incorrect content.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

large, language, models, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of large language models?
2. What practical issue should be considered when applying large language models?
3. How could large language models be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 89: Prompt Engineering

Document ID: AIML-089 | Section: Prompt Engineering

Concept Summary

Prompt engineering structures instructions, context, examples, constraints, and output formats to guide a model toward a desired response.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

prompt, engineering, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of prompt engineering?
2. What practical issue should be considered when applying prompt engineering?
3. How could prompt engineering be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 90: Retrieval Augmented Generation

Document ID: AIML-090 | Section: Retrieval Augmented Generation

Concept Summary

RAG combines retrieval with generation. A retriever selects relevant documents, and a generator uses those documents as context to produce an answer.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

retrieval, augmented, generation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of retrieval augmented generation?
2. What practical issue should be considered when applying retrieval augmented generation?
3. How could retrieval augmented generation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 91: RAG Ingestion

Document ID: AIML-091 | Section: RAG Ingestion

Concept Summary

RAG ingestion loads documents, extracts text, cleans content, adds metadata, and prepares chunks for indexing.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

ingestion, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of rag ingestion?
2. What practical issue should be considered when applying rag ingestion?
3. How could rag ingestion be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 92: Chunking Strategies

Document ID: AIML-092 | Section: Chunking Strategies

Concept Summary

Chunking divides documents into retrievable units. Fixed-size, recursive, semantic, and structure-aware chunking each provide different trade-offs.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

chunking, strategies, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of chunking strategies?
2. What practical issue should be considered when applying chunking strategies?
3. How could chunking strategies be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 93: Vector Databases

Document ID: AIML-093 | Section: Vector Databases

Concept Summary

Vector databases store embeddings and support similarity search. Metadata filtering and indexing strategies improve retrieval relevance.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

vector, databases, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of vector databases?
2. What practical issue should be considered when applying vector databases?
3. How could vector databases be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 94: Embedding Models

Document ID: AIML-094 | Section: Embedding Models

Concept Summary

Embedding models encode text into vectors. Choice of model should consider language coverage, domain, dimensionality, cost, and retrieval quality.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

embedding, models, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of embedding models?
2. What practical issue should be considered when applying embedding models?
3. How could embedding models be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 95: Similarity Search

Document ID: AIML-095 | Section: Similarity Search

Concept Summary

Similarity search ranks vectors according to a distance or similarity measure, such as cosine similarity, dot product, or Euclidean distance.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

similarity, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of similarity search?
2. What practical issue should be considered when applying similarity search?
3. How could similarity search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 96: Hybrid Search

Document ID: AIML-096 | Section: Hybrid Search

Concept Summary

Hybrid search combines lexical retrieval with vector retrieval. It can improve results when exact terms and semantic meaning are both important.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

hybrid, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of hybrid search?
2. What practical issue should be considered when applying hybrid search?
3. How could hybrid search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 97: Reranking

Document ID: AIML-097 | Section: Reranking

Concept Summary

A reranker evaluates retrieved candidates with a more detailed model and reorders them. This can improve precision before generation.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

reranking, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of reranking?
2. What practical issue should be considered when applying reranking?
3. How could reranking be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 98: RAG Evaluation

Document ID: AIML-098 | Section: RAG Evaluation

Concept Summary

RAG evaluation measures retrieval quality, answer relevance, faithfulness, completeness, and latency. Both automated and human evaluation are useful.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

evaluation, model, data, deployment.

RAG Test Questions

1. What is the main idea of rag evaluation?
2. What practical issue should be considered when applying rag evaluation?
3. How could rag evaluation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 99: Hallucination Reduction

Document ID: AIML-099 | Section: Hallucination Reduction

Concept Summary

Grounded context, citation requirements, answer verification, abstention policies, and constrained prompts can reduce unsupported generated claims.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

hallucination, reduction, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of hallucination reduction?
2. What practical issue should be considered when applying hallucination reduction?
3. How could hallucination reduction be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 100: Agentic AI

Document ID: AIML-100 | Section: Agentic AI

Concept Summary

Agentic AI systems plan and execute actions using tools, memory, and feedback. Guardrails are needed to limit unsafe or unintended actions.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

agentic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of agentic ai?
2. What practical issue should be considered when applying agentic ai?
3. How could agentic ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 101: AI Agents and Tools

Document ID: AIML-101 | Section: AI Agents and Tools

Concept Summary

Tool-using agents can call search, databases, calculators, or application APIs. Tool schemas and permission boundaries should be explicit.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

agents, tools, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai agents and tools?
2. What practical issue should be considered when applying ai agents and tools?
3. How could ai agents and tools be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 102: Planning Algorithms

Document ID: AIML-102 | Section: Planning Algorithms

Concept Summary

Planning searches for a sequence of actions that achieves a goal. Classical planning may use symbolic states, while modern agents may combine planning with language models.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

planning, algorithms, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of planning algorithms?
2. What practical issue should be considered when applying planning algorithms?
3. How could planning algorithms be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 103: Robotics

Document ID: AIML-103 | Section: Robotics

Concept Summary

Robotics combines perception, control, planning, and mechanical systems. A robot must handle uncertainty in sensing and interaction with the physical world.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

robotics, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of robotics?
2. What practical issue should be considered when applying robotics?
3. How could robotics be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 104: Computer Vision

Document ID: AIML-104 | Section: Computer Vision

Concept Summary

Computer vision extracts information from images and video. Common operations include filtering, edge detection, feature extraction, recognition, and tracking.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

computer, vision, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of computer vision?
2. What practical issue should be considered when applying computer vision?
3. How could computer vision be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 105: OpenCV Basics

Document ID: AIML-105 | Section: OpenCV Basics

Concept Summary

OpenCV provides tools for reading, writing, transforming, filtering, and analyzing images. Images are commonly represented as multidimensional arrays.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

opencv, basics, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of opencv basics?
2. What practical issue should be considered when applying opencv basics?
3. How could opencv basics be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 106: Image Preprocessing

Document ID: AIML-106 | Section: Image Preprocessing

Concept Summary

Preprocessing can include resizing, denoising, contrast adjustment, normalization, and color-space conversion. Operations should preserve task-relevant information.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

image, preprocessing, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of image preprocessing?
2. What practical issue should be considered when applying image preprocessing?
3. How could image preprocessing be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 107: Speech Recognition

Document ID: AIML-107 | Section: Speech Recognition

Concept Summary

Automatic speech recognition converts audio into text. Challenges include accents, background noise, pronunciation variation, and domain vocabulary.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

speech, recognition, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of speech recognition?
2. What practical issue should be considered when applying speech recognition?
3. How could speech recognition be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 108: Recommendation Systems

Document ID: AIML-108 | Section: Recommendation Systems

Concept Summary

Recommendation systems estimate which items may be useful to a user. Collaborative filtering, content-based methods, and hybrid approaches are common.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

recommendation, systems, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of recommendation systems?
2. What practical issue should be considered when applying recommendation systems?
3. How could recommendation systems be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 109: Time Series Forecasting

Document ID: AIML-109 | Section: Time Series Forecasting

Concept Summary

Time series forecasting predicts future values using historical observations. Seasonality, trend, autocorrelation, and data leakage require careful treatment.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

time, series, forecasting, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of time series forecasting?
2. What practical issue should be considered when applying time series forecasting?
3. How could time series forecasting be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 110: Anomaly Detection

Document ID: AIML-110 | Section: Anomaly Detection

Concept Summary

Anomaly detection identifies observations that differ from expected patterns. Methods include statistical thresholds, isolation forests, autoencoders, and density models.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

anomaly, detection, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of anomaly detection?
2. What practical issue should be considered when applying anomaly detection?
3. How could anomaly detection be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 111: Autoencoders

Document ID: AIML-111 | Section: Autoencoders

Concept Summary

Autoencoders learn to reconstruct input data through an encoded representation. They are used for dimensionality reduction, denoising, and anomaly detection.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

autoencoders, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of autoencoders?
2. What practical issue should be considered when applying autoencoders?
3. How could autoencoders be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 112: Generative Adversarial Networks

Document ID: AIML-112 | Section: Generative Adversarial Networks

Concept Summary

GANs train a generator and discriminator in competition. The generator creates samples while the discriminator distinguishes generated from real examples.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

generative, adversarial, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of generative adversarial networks?
2. What practical issue should be considered when applying generative adversarial networks?
3. How could generative adversarial networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 113: Diffusion Models

Document ID: AIML-113 | Section: Diffusion Models

Concept Summary

Diffusion models learn to reverse a gradual noising process. They are widely used for generating images, audio, and other data types.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

diffusion, models, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of diffusion models?
2. What practical issue should be considered when applying diffusion models?
3. How could diffusion models be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 114: Transfer Learning

Document ID: AIML-114 | Section: Transfer Learning

Concept Summary

Transfer learning reuses knowledge learned from one task or dataset for another. It can reduce data and training requirements.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

transfer, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of transfer learning?
2. What practical issue should be considered when applying transfer learning?
3. How could transfer learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 115: Fine-Tuning

Document ID: AIML-115 | Section: Fine-Tuning

Concept Summary

Fine-tuning adapts a pretrained model using task-specific data. Learning rate, dataset quality, and regularization influence the outcome.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

fine, tuning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of fine-tuning?
2. What practical issue should be considered when applying fine-tuning?
3. How could fine-tuning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 116: Parameter-Efficient Fine-Tuning

Document ID: AIML-116 | Section: Parameter-Efficient Fine-Tuning

Concept Summary

Parameter-efficient methods update a small number of parameters or added modules. Examples include adapters and low-rank adaptation.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

parameter, efficient, fine, tuning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of parameter-efficient fine-tuning?
2. What practical issue should be considered when applying parameter-efficient fine-tuning?
3. How could parameter-efficient fine-tuning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 117: MLOps

Document ID: AIML-117 | Section: MLOps

Concept Summary

MLOps applies engineering practices to the machine learning lifecycle, including versioning, testing, deployment, monitoring, and retraining.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

mlops, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of mlops?
2. What practical issue should be considered when applying mlops?
3. How could mlops be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 118: Model Deployment

Document ID: AIML-118 | Section: Model Deployment

Concept Summary

Deployment makes a trained model available for use through an application, batch job, embedded device, or API.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

model, deployment, data, evaluation.

RAG Test Questions

1. What is the main idea of model deployment?
2. What practical issue should be considered when applying model deployment?
3. How could model deployment be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 119: Model Monitoring

Document ID: AIML-119 | Section: Model Monitoring

Concept Summary

Monitoring tracks performance, latency, resource use, data drift, and failures after deployment. Alerts should connect to actionable response procedures.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

model, monitoring, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of model monitoring?
2. What practical issue should be considered when applying model monitoring?
3. How could model monitoring be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 120: Data Drift

Document ID: AIML-120 | Section: Data Drift

Concept Summary

Data drift occurs when the distribution of inputs changes over time. Drift detection can compare summary statistics, embeddings, or model-specific signals.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

data, drift, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of data drift?
2. What practical issue should be considered when applying data drift?
3. How could data drift be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 121: Concept Drift

Document ID: AIML-121 | Section: Concept Drift

Concept Summary

Concept drift occurs when the relationship between inputs and targets changes. It may require retraining, recalibration, or changes to the modeling approach.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

concept, drift, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of concept drift?
2. What practical issue should be considered when applying concept drift?
3. How could concept drift be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 122: Reproducibility

Document ID: AIML-122 | Section: Reproducibility

Concept Summary

Reproducibility requires recording data versions, code, configurations, random seeds, dependencies, and evaluation procedures.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

reproducibility, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of reproducibility?
2. What practical issue should be considered when applying reproducibility?
3. How could reproducibility be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 123: AI Project Lifecycle

Document ID: AIML-123 | Section: AI Project Lifecycle

Concept Summary

A practical AI project moves from problem definition to data collection, exploration, modeling, evaluation, deployment, and monitoring.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

project, lifecycle, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai project lifecycle?
2. What practical issue should be considered when applying ai project lifecycle?
3. How could ai project lifecycle be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 124: Problem Framing

Document ID: AIML-124 | Section: Problem Framing

Concept Summary

Problem framing clarifies the user, decision, target, constraints, success metrics, and consequences of errors before selecting an algorithm.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

problem, framing, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of problem framing?
2. What practical issue should be considered when applying problem framing?
3. How could problem framing be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 125: Data Collection

Document ID: AIML-125 | Section: Data Collection

Concept Summary

Data collection should address relevance, representativeness, consent, quality, provenance, and legal or ethical constraints.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

data, collection, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of data collection?
2. What practical issue should be considered when applying data collection?
3. How could data collection be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 126: Data Annotation

Document ID: AIML-126 | Section: Data Annotation

Concept Summary

Annotation creates labels or structured information. Clear guidelines, reviewer training, and agreement checks improve label consistency.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

data, annotation, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of data annotation?
2. What practical issue should be considered when applying data annotation?
3. How could data annotation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 127: Experimental Design

Document ID: AIML-127 | Section: Experimental Design

Concept Summary

A strong experiment changes one major factor at a time when possible, uses appropriate baselines, and reports uncertainty and limitations.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

experimental, design, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of experimental design?
2. What practical issue should be considered when applying experimental design?
3. How could experimental design be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 128: Baseline Models

Document ID: AIML-128 | Section: Baseline Models

Concept Summary

A baseline provides a reference point for evaluating more complex approaches. Simple rules and traditional models are often valuable baselines.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

baseline, models, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of baseline models?
2. What practical issue should be considered when applying baseline models?
3. How could baseline models be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 129: Statistical Significance

Document ID: AIML-129 | Section: Statistical Significance

Concept Summary

Statistical significance evaluates whether an observed difference is compatible with a specified null hypothesis. Practical significance and uncertainty should also be considered.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

statistical, significance, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of statistical significance?
2. What practical issue should be considered when applying statistical significance?
3. How could statistical significance be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 130: Ablation Studies

Document ID: AIML-130 | Section: Ablation Studies

Concept Summary

Ablation studies remove or alter components to estimate their contribution. They help identify which parts of a system actually improve results.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

ablation, studies, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ablation studies?
2. What practical issue should be considered when applying ablation studies?
3. How could ablation studies be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 131: Error Analysis

Document ID: AIML-131 | Section: Error Analysis

Concept Summary

Error analysis groups failures by cause, population, input type, or severity. It guides targeted improvements more effectively than a single aggregate metric.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

error, analysis, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of error analysis?
2. What practical issue should be considered when applying error analysis?
3. How could error analysis be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 132: Responsible AI Documentation

Document ID: AIML-132 | Section: Responsible AI Documentation

Concept Summary

Documentation should describe intended use, limitations, data sources, evaluation groups, known risks, and monitoring plans.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

responsible, documentation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of responsible ai documentation?
2. What practical issue should be considered when applying responsible ai documentation?
3. How could responsible ai documentation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 133: AI Governance

Document ID: AIML-133 | Section: AI Governance

Concept Summary

AI governance defines roles, policies, review processes, accountability, and controls for developing and operating AI systems.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

governance, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai governance?
2. What practical issue should be considered when applying ai governance?
3. How could ai governance be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 134: Security of ML Systems

Document ID: AIML-134 | Section: Security of ML Systems

Concept Summary

ML security includes protection against data poisoning, model theft, prompt injection, adversarial examples, and unauthorized access.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

security, systems, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of security of ml systems?
2. What practical issue should be considered when applying security of ml systems?
3. How could security of ml systems be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 135: Prompt Injection

Document ID: AIML-135 | Section: Prompt Injection

Concept Summary

Prompt injection attempts to manipulate a language model through untrusted instructions in user input or retrieved content. Systems should separate data from trusted commands.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

prompt, injection, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of prompt injection?
2. What practical issue should be considered when applying prompt injection?
3. How could prompt injection be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 136: Adversarial Examples

Document ID: AIML-136 | Section: Adversarial Examples

Concept Summary

Adversarial examples are inputs modified to cause incorrect model behavior. Robustness testing and defensive strategies can reveal and reduce vulnerabilities.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

adversarial, examples, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of adversarial examples?
2. What practical issue should be considered when applying adversarial examples?
3. How could adversarial examples be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 137: Federated Learning

Document ID: AIML-137 | Section: Federated Learning

Concept Summary

Federated learning trains models across decentralized devices or organizations while sharing selected updates rather than raw data.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

federated, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of federated learning?
2. What practical issue should be considered when applying federated learning?
3. How could federated learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 138: Differential Privacy

Document ID: AIML-138 | Section: Differential Privacy

Concept Summary

Differential privacy limits how much the output of an analysis reveals about any individual record. Privacy budgets quantify a trade-off between privacy and utility.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

differential, privacy, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of differential privacy?
2. What practical issue should be considered when applying differential privacy?
3. How could differential privacy be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 139: Edge AI

Document ID: AIML-139 | Section: Edge AI

Concept Summary

Edge AI runs inference close to where data is generated. Benefits may include lower latency, reduced bandwidth use, and improved local control.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

edge, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of edge ai?
2. What practical issue should be considered when applying edge ai?
3. How could edge ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 140: AI in Healthcare

Document ID: AIML-140 | Section: AI in Healthcare

Concept Summary

Healthcare AI may support imaging, triage, documentation, and research. Validation, privacy, clinical oversight, and subgroup evaluation are essential.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

healthcare, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in healthcare?
2. What practical issue should be considered when applying ai in healthcare?
3. How could ai in healthcare be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 141: AI in Education

Document ID: AIML-141 | Section: AI in Education

Concept Summary

Education AI can support tutoring, feedback, content generation, and learning analytics. Transparency and teacher oversight are important.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

education, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in education?
2. What practical issue should be considered when applying ai in education?
3. How could ai in education be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 142: AI in Finance

Document ID: AIML-142 | Section: AI in Finance

Concept Summary

Finance applications include fraud detection, forecasting, risk analysis, and customer support. Explainability, security, and regulatory obligations matter.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

finance, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in finance?
2. What practical issue should be considered when applying ai in finance?
3. How could ai in finance be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 143: AI in Agriculture

Document ID: AIML-143 | Section: AI in Agriculture

Concept Summary

Agricultural AI can analyze crop images, weather patterns, soil information, and yield data. Field conditions and regional variation affect reliability.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

agriculture, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in agriculture?
2. What practical issue should be considered when applying ai in agriculture?
3. How could ai in agriculture be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 144: AI in E-Commerce

Document ID: AIML-144 | Section: AI in E-Commerce

Concept Summary

E-commerce AI supports search, recommendations, demand forecasting, customer service, and fraud detection.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

commerce, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in e-commerce?
2. What practical issue should be considered when applying ai in e-commerce?
3. How could ai in e-commerce be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 145: AI in Cybersecurity

Document ID: AIML-145 | Section: AI in Cybersecurity

Concept Summary

Cybersecurity AI detects suspicious behavior, prioritizes alerts, and assists analysts. Attackers may adapt, so human review and layered controls remain important.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

cybersecurity, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in cybersecurity?
2. What practical issue should be considered when applying ai in cybersecurity?
3. How could ai in cybersecurity be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 146: AI in Smart Cities

Document ID: AIML-146 | Section: AI in Smart Cities

Concept Summary

Smart city systems may optimize traffic, energy, waste, and public services. Governance should address surveillance risks and equitable access.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

smart, cities, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai in smart cities?
2. What practical issue should be considered when applying ai in smart cities?
3. How could ai in smart cities be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 147: Multimodal AI

Document ID: AIML-147 | Section: Multimodal AI

Concept Summary

Multimodal systems combine information from text, images, audio, video, or sensor data. Alignment across modalities is a major design challenge.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

multimodal, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of multimodal ai?
2. What practical issue should be considered when applying multimodal ai?
3. How could multimodal ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 148: Knowledge Graphs

Document ID: AIML-148 | Section: Knowledge Graphs

Concept Summary

Knowledge graphs represent entities and relationships as connected structures. They can support search, reasoning, recommendations, and retrieval grounding.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

knowledge, graphs, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of knowledge graphs?
2. What practical issue should be considered when applying knowledge graphs?
3. How could knowledge graphs be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 149: Graph Neural Networks

Document ID: AIML-149 | Section: Graph Neural Networks

Concept Summary

Graph neural networks propagate information among connected nodes. They are useful for molecules, social networks, recommendation graphs, and relational data.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

graph, neural, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of graph neural networks?
2. What practical issue should be considered when applying graph neural networks?
3. How could graph neural networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 150: Reinforcement Learning from Feedback

Document ID: AIML-150 | Section: Reinforcement Learning from Feedback

Concept Summary

Feedback-based training uses preferences or evaluations to shape model behavior. Feedback quality, reward design, and unintended optimization require attention.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

reinforcement, learning, from, feedback, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of reinforcement learning from feedback?
2. What practical issue should be considered when applying reinforcement learning from feedback?
3. How could reinforcement learning from feedback be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 151: Evaluation Data Design

Document ID: AIML-151 | Section: Evaluation Data Design

Concept Summary

Evaluation sets should represent real use cases, difficult cases, edge cases, and safety-sensitive situations. Test contamination should be considered.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

evaluation, data, design, model, deployment.

RAG Test Questions

1. What is the main idea of evaluation data design?
2. What practical issue should be considered when applying evaluation data design?
3. How could evaluation data design be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 152: Synthetic Data

Document ID: AIML-152 | Section: Synthetic Data

Concept Summary

Synthetic data is generated rather than directly collected. It can increase coverage, but errors, bias, and unrealistic patterns may be inherited or introduced.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

synthetic, data, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of synthetic data?
2. What practical issue should be considered when applying synthetic data?
3. How could synthetic data be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 153: Data-Centric AI

Document ID: AIML-153 | Section: Data-Centric AI

Concept Summary

Data-centric AI focuses on improving data quality, labels, coverage, and consistency rather than only changing model architecture.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

data, centric, model, evaluation, deployment.

RAG Test Questions

1. What is the main idea of data-centric ai?
2. What practical issue should be considered when applying data-centric ai?
3. How could data-centric ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 154: Model-Centric AI

Document ID: AIML-154 | Section: Model-Centric AI

Concept Summary

Model-centric improvement changes architectures, optimization, or training methods while holding the data process relatively stable.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

model, centric, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of model-centric ai?
2. What practical issue should be considered when applying model-centric ai?
3. How could model-centric ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 155: Uncertainty Estimation

Document ID: AIML-155 | Section: Uncertainty Estimation

Concept Summary

Uncertainty estimation communicates how confident a model is. Predictive uncertainty can arise from noisy data, limited coverage, or model limitations.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

uncertainty, estimation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of uncertainty estimation?
2. What practical issue should be considered when applying uncertainty estimation?
3. How could uncertainty estimation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 156: Out-of-Distribution Detection

Document ID: AIML-156 | Section: Out-of-Distribution Detection

Concept Summary

Out-of-distribution detection identifies inputs that differ from the training distribution. It can support safer abstention and human review.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

distribution, detection, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of out-of-distribution detection?
2. What practical issue should be considered when applying out-of-distribution detection?
3. How could out-of-distribution detection be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 157: Continual Learning

Document ID: AIML-157 | Section: Continual Learning

Concept Summary

Continual learning updates a model as new data arrives. Catastrophic forgetting and changing task requirements must be managed.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

continual, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of continual learning?
2. What practical issue should be considered when applying continual learning?
3. How could continual learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 158: Active Learning

Document ID: AIML-158 | Section: Active Learning

Concept Summary

Active learning selects the most informative examples for labeling. It can reduce annotation costs when model uncertainty or diversity is used effectively.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

active, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of active learning?
2. What practical issue should be considered when applying active learning?
3. How could active learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 159: Human-in-the-Loop AI

Document ID: AIML-159 | Section: Human-in-the-Loop AI

Concept Summary

Human-in-the-loop systems assign appropriate decisions to people and models. Interfaces should make uncertainty, evidence, and escalation pathways clear.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

human, loop, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of human-in-the-loop ai?
2. What practical issue should be considered when applying human-in-the-loop ai?
3. How could human-in-the-loop ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 160: AI Usability

Document ID: AIML-160 | Section: AI Usability

Concept Summary

An AI tool must fit the user's workflow. Usability includes understandable outputs, reasonable response time, recoverability, accessibility, and feedback mechanisms.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

usability, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai usability?
2. What practical issue should be considered when applying ai usability?
3. How could ai usability be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 161: AI System Testing

Document ID: AIML-161 | Section: AI System Testing

Concept Summary

Testing can include unit tests, integration tests, data validation, performance tests, adversarial tests, and user acceptance tests.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

system, testing, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai system testing?
2. What practical issue should be considered when applying ai system testing?
3. How could ai system testing be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 162: API Design for AI

Document ID: AIML-162 | Section: API Design for AI

Concept Summary

An AI API should define input schemas, output schemas, errors, authentication, rate limits, timeouts, and versioning.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

design, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of api design for ai?
2. What practical issue should be considered when applying api design for ai?
3. How could api design for ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 163: Caching and Latency

Document ID: AIML-163 | Section: Caching and Latency

Concept Summary

Caching repeated computations can reduce cost and response time. Latency budgets should account for retrieval, model inference, network calls, and post-processing.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

caching, latency, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of caching and latency?
2. What practical issue should be considered when applying caching and latency?
3. How could caching and latency be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 164: Cost Optimization

Document ID: AIML-164 | Section: Cost Optimization

Concept Summary

AI cost optimization may involve smaller models, batching, caching, quantization, retrieval filtering, and selective use of expensive operations.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

cost, optimization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of cost optimization?
2. What practical issue should be considered when applying cost optimization?
3. How could cost optimization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 165: Quantization

Document ID: AIML-165 | Section: Quantization

Concept Summary

Quantization represents model values with lower numerical precision. It can reduce memory and speed inference but may affect accuracy.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

quantization, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of quantization?
2. What practical issue should be considered when applying quantization?
3. How could quantization be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 166: Model Compression

Document ID: AIML-166 | Section: Model Compression

Concept Summary

Compression techniques include pruning, distillation, quantization, and low-rank representations. The best method depends on hardware and quality requirements.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

model, compression, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of model compression?
2. What practical issue should be considered when applying model compression?
3. How could model compression be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 167: Knowledge Distillation

Document ID: AIML-167 | Section: Knowledge Distillation

Concept Summary

Knowledge distillation trains a smaller student model to reproduce useful behavior from a larger teacher model.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

knowledge, distillation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of knowledge distillation?
2. What practical issue should be considered when applying knowledge distillation?
3. How could knowledge distillation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 168: Ensemble Learning

Document ID: AIML-168 | Section: Ensemble Learning

Concept Summary

Ensembles combine predictions from multiple models. Diversity among component errors is often important for obtaining gains.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

ensemble, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ensemble learning?
2. What practical issue should be considered when applying ensemble learning?
3. How could ensemble learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 169: Stacking and Blending

Document ID: AIML-169 | Section: Stacking and Blending

Concept Summary

Stacking learns how to combine base model outputs using a meta-model. Blending combines predictions using a held-out dataset or fixed weights.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

stacking, blending, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of stacking and blending?
2. What practical issue should be considered when applying stacking and blending?
3. How could stacking and blending be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 170: Online Learning

Document ID: AIML-170 | Section: Online Learning

Concept Summary

Online learning updates a model incrementally as new examples arrive. It requires safeguards against noisy or malicious updates.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

online, learning, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of online learning?
2. What practical issue should be considered when applying online learning?
3. How could online learning be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 171: Active Monitoring Dashboards

Document ID: AIML-171 | Section: Active Monitoring Dashboards

Concept Summary

Monitoring dashboards should show key metrics, trends, segment breakdowns, alert status, and links to investigation details.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

active, monitoring, dashboards, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of active monitoring dashboards?
2. What practical issue should be considered when applying active monitoring dashboards?
3. How could active monitoring dashboards be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 172: Incident Response for AI

Document ID: AIML-172 | Section: Incident Response for AI

Concept Summary

AI incident response includes detection, triage, containment, rollback, communication, root-cause analysis, and preventive action.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

incident, response, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of incident response for ai?
2. What practical issue should be considered when applying incident response for ai?
3. How could incident response for ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 173: AI Documentation Templates

Document ID: AIML-173 | Section: AI Documentation Templates

Concept Summary

A useful model document includes purpose, inputs, outputs, training data, metrics, limitations, risks, intended users, and maintenance expectations.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

documentation, templates, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of ai documentation templates?
2. What practical issue should be considered when applying ai documentation templates?
3. How could ai documentation templates be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 174: RAG Security

Document ID: AIML-174 | Section: RAG Security

Concept Summary

RAG systems must treat retrieved documents as potentially untrusted. Access permissions, source filtering, prompt-injection defenses, and citation checks are important.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

security, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of rag security?
2. What practical issue should be considered when applying rag security?
3. How could rag security be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 175: RAG with Metadata

Document ID: AIML-175 | Section: RAG with Metadata

Concept Summary

Metadata such as source, date, author, department, and access level can improve filtering, traceability, and access control in retrieval systems.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

with, metadata, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of rag with metadata?
2. What practical issue should be considered when applying rag with metadata?
3. How could rag with metadata be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 176: RAG Query Rewriting

Document ID: AIML-176 | Section: RAG Query Rewriting

Concept Summary

Query rewriting transforms an unclear user question into a retrieval-friendly query. The rewritten query should preserve the user's intent and constraints.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

query, rewriting, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of rag query rewriting?
2. What practical issue should be considered when applying rag query rewriting?
3. How could rag query rewriting be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 177: Multi-Query Retrieval

Document ID: AIML-177 | Section: Multi-Query Retrieval

Concept Summary

Multi-query retrieval generates alternative formulations of a question and combines results. It can improve recall but may increase cost and noise.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

multi, query, retrieval, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of multi-query retrieval?
2. What practical issue should be considered when applying multi-query retrieval?
3. How could multi-query retrieval be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 178: Context Compression

Document ID: AIML-178 | Section: Context Compression

Concept Summary

Context compression removes irrelevant passages or summarizes retrieved material before generation. It helps fit useful evidence within a context window.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

context, compression, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of context compression?
2. What practical issue should be considered when applying context compression?
3. How could context compression be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 179: Citation Generation

Document ID: AIML-179 | Section: Citation Generation

Concept Summary

Citation generation links answer claims to supporting sources. Citations should be accurate, relevant, and placed near the claims they support.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

citation, generation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of citation generation?
2. What practical issue should be considered when applying citation generation?
3. How could citation generation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 180: RAG Failure Modes

Document ID: AIML-180 | Section: RAG Failure Modes

Concept Summary

Common RAG failures include poor chunking, missing documents, irrelevant retrieval, conflicting sources, context overload, and unsupported synthesis.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

failure, modes, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of rag failure modes?
2. What practical issue should be considered when applying rag failure modes?
3. How could rag failure modes be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 181: RAG Benchmarking

Document ID: AIML-181 | Section: RAG Benchmarking

Concept Summary

A RAG benchmark should define representative questions, expected evidence, retrieval metrics, answer metrics, latency, cost, and failure categories.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

benchmarking, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of rag benchmarking?
2. What practical issue should be considered when applying rag benchmarking?
3. How could rag benchmarking be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 182: Final AI and ML Revision

Document ID: AIML-182 | Section: Final AI and ML Revision

Concept Summary

A complete AI/ML workflow connects problem framing, data, algorithms, evaluation, deployment, monitoring, and responsible use. Strong systems are measured against real objectives rather than novelty alone.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

final, revision, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of final ai and ml revision?
2. What practical issue should be considered when applying final ai and ml revision?
3. How could final ai and ml revision be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 183: Foundations of Artificial Intelligence

Document ID: AIML-183 | Section: Foundations of Artificial Intelligence

Concept Summary

AI is the field of building systems that perform tasks associated with human intelligence, including perception, reasoning, learning, planning, language understanding, and decision-making.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

foundations, artificial, intelligence, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of foundations of artificial intelligence?
2. What practical issue should be considered when applying foundations of artificial intelligence?
3. How could foundations of artificial intelligence be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 184: History of Artificial Intelligence

Document ID: AIML-184 | Section: History of Artificial Intelligence

Concept Summary

AI developed through several periods: symbolic reasoning, expert systems, statistical learning, deep learning, and foundation models. Progress has been influenced by algorithms, data, computing power, and evaluation methods.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

history, artificial, intelligence, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of history of artificial intelligence?
2. What practical issue should be considered when applying history of artificial intelligence?
3. How could history of artificial intelligence be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 185: Intelligent Agents

Document ID: AIML-185 | Section: Intelligent Agents

Concept Summary

An intelligent agent perceives its environment through sensors and acts through actuators. A rational agent chooses actions that maximize expected performance according to its objective and available information.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

intelligent, agents, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of intelligent agents?
2. What practical issue should be considered when applying intelligent agents?
3. How could intelligent agents be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 186: Problem Solving in AI

Document ID: AIML-186 | Section: Problem Solving in AI

Concept Summary

Many AI tasks can be expressed as a state-space search problem. A state describes a situation, actions create transitions, and a goal test determines whether a solution has been reached.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

problem, solving, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of problem solving in ai?
2. What practical issue should be considered when applying problem solving in ai?
3. How could problem solving in ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 187: Breadth-First Search

Document ID: AIML-187 | Section: Breadth-First Search

Concept Summary

Breadth-first search expands the shallowest unexplored nodes first. With unit step costs, it is complete and returns a shortest path when the branching factor is finite.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

breadth, first, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of breadth-first search?
2. What practical issue should be considered when applying breadth-first search?
3. How could breadth-first search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 188: Depth-First Search

Document ID: AIML-188 | Section: Depth-First Search

Concept Summary

Depth-first search explores one branch as deeply as possible before backtracking. It uses less memory than breadth-first search but may become trapped in deep or infinite branches.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

depth, first, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of depth-first search?
2. What practical issue should be considered when applying depth-first search?
3. How could depth-first search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 189: Uniform-Cost Search

Document ID: AIML-189 | Section: Uniform-Cost Search

Concept Summary

Uniform-cost search expands the node with the lowest path cost. It is optimal when step costs are nonnegative and the queue is ordered by cumulative path cost.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

uniform, cost, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of uniform-cost search?
2. What practical issue should be considered when applying uniform-cost search?
3. How could uniform-cost search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 190: A-Star Search

Document ID: AIML-190 | Section: A-Star Search

Concept Summary

A-star search combines the path cost so far, $g(n)$, with a heuristic estimate, $h(n)$. The evaluation function is $f(n)=g(n)+h(n)$. An admissible heuristic never overestimates the remaining cost.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

star, search, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of a-star search?
2. What practical issue should be considered when applying a-star search?
3. How could a-star search be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 191: Heuristic Functions

Document ID: AIML-191 | Section: Heuristic Functions

Concept Summary

A heuristic guides search toward promising states. Good heuristics reduce computation while preserving solution quality when their properties are understood.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

heuristic, functions, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of heuristic functions?
2. What practical issue should be considered when applying heuristic functions?
3. How could heuristic functions be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 192: Knowledge Representation

Document ID: AIML-192 | Section: Knowledge Representation

Concept Summary

Knowledge representation encodes facts, relationships, rules, and uncertainty in a form that an AI system can use for inference. Common approaches include logic, semantic networks, frames, and ontologies.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

knowledge, representation, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of knowledge representation?
2. What practical issue should be considered when applying knowledge representation?
3. How could knowledge representation be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 193: Propositional Logic

Document ID: AIML-193 | Section: Propositional Logic

Concept Summary

Propositional logic represents statements that are either true or false. Logical connectives include AND, OR, NOT, implication, and biconditional.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

propositional, logic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of propositional logic?
2. What practical issue should be considered when applying propositional logic?
3. How could propositional logic be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 194: First-Order Logic

Document ID: AIML-194 | Section: First-Order Logic

Concept Summary

First-order logic extends propositional logic with variables, predicates, functions, and quantifiers. It can represent relationships among objects and general rules.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

first, order, logic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of first-order logic?
2. What practical issue should be considered when applying first-order logic?
3. How could first-order logic be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 195: Expert Systems

Document ID: AIML-195 | Section: Expert Systems

Concept Summary

An expert system typically includes a knowledge base and an inference engine. It applies domain rules to known facts and may explain how a conclusion was derived.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

expert, systems, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of expert systems?
2. What practical issue should be considered when applying expert systems?
3. How could expert systems be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 196: Fuzzy Logic

Document ID: AIML-196 | Section: Fuzzy Logic

Concept Summary

Fuzzy logic represents degrees of membership rather than only binary membership. It is useful when concepts such as hot, near, or fast have gradual boundaries.

Learning Note

Key idea: Start with a clear objective and define what success means before choosing a technique.

Important Terms

fuzzy, logic, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of fuzzy logic?
2. What practical issue should be considered when applying fuzzy logic?
3. How could fuzzy logic be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 197: Probability in AI

Document ID: AIML-197 | Section: Probability in AI

Concept Summary

Probability provides a mathematical language for uncertainty. Conditional probability, Bayes' theorem, and probability distributions are central to probabilistic reasoning.

Learning Note

Practical perspective: Use a small baseline first, then compare more complex methods under the same evaluation protocol.

Important Terms

probability, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of probability in ai?
2. What practical issue should be considered when applying probability in ai?
3. How could probability in ai be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 198: Bayesian Networks

Document ID: AIML-198 | Section: Bayesian Networks

Concept Summary

A Bayesian network is a directed acyclic graph whose nodes represent random variables and whose edges represent conditional dependencies. It factorizes a joint probability distribution.

Learning Note

Implementation note: Check data quality, leakage, and representativeness before trusting a numerical result.

Important Terms

bayesian, networks, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of bayesian networks?
2. What practical issue should be considered when applying bayesian networks?
3. How could bayesian networks be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 199: Decision Trees

Document ID: AIML-199 | Section: Decision Trees

Concept Summary

A decision tree predicts a target by applying a sequence of feature-based tests. Splits may be selected using information gain, gain ratio, or Gini impurity.

Learning Note

Evaluation perspective: Document assumptions, limitations, and the conditions under which the method may fail.

Important Terms

decision, trees, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of decision trees?
2. What practical issue should be considered when applying decision trees?
3. How could decision trees be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.

Page 200: Machine Learning Overview

Document ID: AIML-200 | Section: Machine Learning Overview

Concept Summary

Machine learning enables a system to learn patterns from examples rather than relying entirely on manually written rules. The learning process includes data preparation, training, validation, and testing.

Learning Note

Common mistake: Separate training, validation, and testing so that reported performance remains credible.

Important Terms

machine, learning, overview, model, data, evaluation, deployment.

RAG Test Questions

1. What is the main idea of machine learning overview?
2. What practical issue should be considered when applying machine learning overview?
3. How could machine learning overview be evaluated in an AI or ML project?

Short Answer Cue

Answer using the concept summary, learning note, and the specific constraints of the question. If evidence is missing, state the limitation instead of inventing details.